

3.39. SOLUTION OF A SYSTEM OF LINEAR EQUATIONS

Consider the system of equations
$$\left. \begin{aligned} a_1x + b_1y + c_1z &= d_1 \\ a_2x + b_2y + c_2z &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned} \right\} \text{ (3 equations in 3 unknowns)}$$

In matrix notation, these equations can be written as

$$\begin{bmatrix} a_1x + b_1y + c_1z \\ a_2x + b_2y + c_2z \\ a_3x + b_3y + c_3z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

or

$$\begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

or

$$AX = B$$

where $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ is called the co-efficient matrix, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is the column matrix of

unknowns, $B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$ is the column matrix of constants.

If $d_1 = d_2 = d_3 = 0$, then $B = O$ and the matrix equation $AX = B$ reduces to $AX = O$.

Such a system of equations is called a system of **homogeneous linear equations**.

If atleast one of d_1, d_2, d_3 is non-zero, then $B \neq O$.

Such a system of equations is called a system of **non-homogeneous linear equations**.

Solving the matrix equation $AX = B$ means finding X , i.e., finding a column matrix $\begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$ such that

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}. \text{ Then } x = \alpha, y = \beta, z = \gamma.$$

The matrix equation $AX = B$ need not always have a solution. It may have no solution or a unique solution or an infinite number of solutions.

A system of equations having **no solution** is called an **inconsistent** system of equations.

A system of equations having **one or more solution** is called a **consistent** system of equations.

For a system of non-homogeneous linear equations $AX = B$.

- (i) if $\rho[A : B] \neq \rho(A)$, the system is inconsistent.
 - (ii) if $\rho[A : B] = \rho(A) = \text{number of unknowns}$, the system has a unique solution.
 - (iii) $\rho[A : B] = \rho(A) < \text{number of unknowns}$, the system has an infinite number of solutions.
- The matrix $[A : B]$ in which the elements of A and B are written side by side is called the **augmented matrix**.

For a system of homogeneous linear equations $AX = O$

(i) $X = O$ is always a solution. This solution in which each unknown has the value zero is called the **Null Solution** or the **Trivial Solution**. Thus a homogeneous system is always consistent.

A system of homogeneous linear equations has either the trivial solution or an infinite number of solutions.

- (ii) if $\rho(A) = \text{number of unknowns}$, the system has only the trivial solution.
- (iii) if $\rho(A) < \text{number of unknowns}$, the system has an infinite number of non-trivial solutions.

3.40. IF A IS A NON-SINGULAR MATRIX, THEN THE MATRIX EQUATION $AX = B$ HAS A UNIQUE SOLUTION

The given equation is $AX = B$...(1)

$\therefore A$ is a non-singular matrix, $\therefore A^{-1}$ exists.

Pre-multiplying both sides of (1) by A^{-1} , we get

$$A^{-1}AX = A^{-1}B \quad \text{or} \quad (A^{-1}A)X = A^{-1}B$$

$$\text{or} \quad IX = A^{-1}B \quad \text{or} \quad X = A^{-1}B$$

which is the required unique solution (since A^{-1} is unique).

Another Method

Write the augmented matrix $[A : B]$. By **E-row operations** on A and B , reduce A to a diagonal matrix, thus getting

$$[A : B] \sim \begin{bmatrix} p_1 & 0 & 0 & \vdots & q_1 \\ 0 & p_2 & 0 & \vdots & q_2 \\ 0 & 0 & p_3 & \vdots & q_3 \end{bmatrix}$$

$$\text{Then} \quad p_1x = q_1, \quad p_2y = q_2, \quad p_3z = q_3.$$

ILLUSTRATIVE EXAMPLES

Example 1. Solve the system of equations :

$$5x + 3y + 7z = 4, \quad 3x + 26y + 2z = 9, \quad 7x + 2y + 11z = 5$$

with the help of matrix inversion.

Sol. In matrix notation, the given system of equations can be written as

$$AX = B$$

$$\text{where } A = \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 11 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix}$$

...(1)

Let $A = \begin{bmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 11 \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$

$$|A| = \begin{vmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 11 \end{vmatrix} = 5 \begin{vmatrix} 26 & 2 \\ 2 & 11 \end{vmatrix} - 3 \begin{vmatrix} 3 & 2 \\ 7 & 11 \end{vmatrix} + 7 \begin{vmatrix} 3 & 26 \\ 7 & 2 \end{vmatrix}$$

$$= 5(286 - 4) - 3(33 - 14) + 7(6 - 182) = 1410 - 57 - 1232 = 121 \neq 0$$

$\Rightarrow A$ is non-singular $\therefore A^{-1}$ exists and the unique solution of (1) is

$$X = A^{-1}B$$

...(2)

$$A_1 = \begin{vmatrix} 26 & 2 \\ 2 & 11 \end{vmatrix} = 282, \quad A_2 = - \begin{vmatrix} 3 & 7 \\ 2 & 11 \end{vmatrix} = -19, \quad A_3 = \begin{vmatrix} 3 & 7 \\ 26 & 2 \end{vmatrix} = -176$$

$$B_1 = - \begin{vmatrix} 3 & 2 \\ 7 & 11 \end{vmatrix} = -19, \quad B_2 = \begin{vmatrix} 5 & 7 \\ 7 & 11 \end{vmatrix} = 6, \quad B_3 = - \begin{vmatrix} 5 & 7 \\ 3 & 2 \end{vmatrix} = 11$$

$$C_1 = \begin{vmatrix} 3 & 26 \\ 7 & 2 \end{vmatrix} = -176, \quad C_2 = - \begin{vmatrix} 5 & 3 \\ 7 & 2 \end{vmatrix} = 11, \quad C_3 = \begin{vmatrix} 5 & 3 \\ 3 & 26 \end{vmatrix} = 121$$

$$\therefore \text{adj } A = \text{transpose of } \begin{bmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{bmatrix} = \begin{bmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{bmatrix} = \begin{bmatrix} 282 & -19 & -176 \\ -19 & 6 & 11 \\ -176 & 11 & 121 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{121} \begin{bmatrix} 282 & -19 & -176 \\ -19 & 6 & 11 \\ -176 & 11 & 121 \end{bmatrix}$$

From (2), $X = A^{-1}B = \frac{1}{121} \begin{bmatrix} 282 & -19 & -176 \\ -19 & 6 & 11 \\ -176 & 11 & 121 \end{bmatrix} \begin{bmatrix} 4 \\ 9 \\ 5 \end{bmatrix} = \frac{1}{121} \begin{bmatrix} 282(4) - 19(9) - 176(5) \\ -19(4) + 6(9) + 11(5) \\ -176(4) + 11(9) + 121(5) \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{121} \begin{bmatrix} 77 \\ 33 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{77}{121} \\ \frac{33}{121} \\ \frac{0}{121} \end{bmatrix} = \begin{bmatrix} \frac{7}{11} \\ \frac{3}{11} \\ 0 \end{bmatrix}$$

Hence $x = \frac{7}{11}, y = \frac{3}{11}, z = 0.$

Example 2. Solve, with the help of matrices, the simultaneous equations :

$$x + y + z = 3, \quad x + 2y + 3z = 4, \quad x + 4y + 9z = 6.$$

Sol. [In this question, there is no restriction that the solution must be obtained by finding A^{-1} .]

Augmented matrix $[A : B] = \begin{bmatrix} 1 & 1 & 1 & \vdots & 3 \\ 1 & 2 & 3 & \vdots & 4 \\ 1 & 4 & 9 & \vdots & 6 \end{bmatrix}$

Operating $R_2 - R_1, R_3 - R_1$

$$= \begin{bmatrix} 1 & 1 & 1 & \vdots & 3 \\ 0 & 1 & 2 & \vdots & 1 \\ 0 & 3 & 8 & \vdots & 3 \end{bmatrix} \text{ Operating } R_3 - 3R_2$$

$$- \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & 2 & : & 1 \\ 0 & 0 & 2 & : & 0 \end{bmatrix} \text{ Operating } (\frac{1}{2})R_3$$

$$- \begin{bmatrix} 1 & 1 & 1 & : & 3 \\ 0 & 1 & 2 & : & 1 \\ 0 & 0 & 1 & : & 0 \end{bmatrix} \text{ Operating } R_1 - R_3, R_2 - 2R_3$$

$$- \begin{bmatrix} 1 & 1 & 0 & : & 3 \\ 0 & 1 & 0 & : & 1 \\ 0 & 0 & 1 & : & 0 \end{bmatrix} \text{ Operating } R_1 - R_2$$

$$- \begin{bmatrix} 1 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & : & 1 \\ 0 & 0 & 1 & : & 0 \end{bmatrix}$$

$$\therefore x = 2, y = 1, z = 0.$$

[This method is especially useful when the number of unknowns is 4, since $|A|$ is of order 4 and the co-factors of its various elements are determinants of order 3.]

Example 3. Solve the system of equations :

$$2x_1 + x_2 + 2x_3 + x_4 = 6, \quad 6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1, \quad 2x_1 + 2x_2 - x_3 + x_4 = 10.$$

Sol. In matrix notation, the given system of equations can be written as $AX = B$

$$\text{where } A = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 6 & -6 & 6 & 12 \\ 4 & 3 & 3 & -3 \\ 2 & 2 & -1 & 1 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, B = \begin{bmatrix} 6 \\ 36 \\ -1 \\ 10 \end{bmatrix}$$

Augmented matrix $[A : B]$

$$= \begin{bmatrix} 2 & 1 & 2 & 1 & : & 6 \\ 6 & -6 & 6 & 12 & : & 36 \\ 4 & 3 & 3 & -3 & : & -1 \\ 2 & 2 & -1 & 1 & : & 10 \end{bmatrix} \text{ Operating } R_2 - 3R_1, R_3 - 2R_1, R_4 - R_1$$

$$- \begin{bmatrix} 2 & 1 & 2 & 1 & : & 6 \\ 0 & -9 & 0 & 9 & : & 18 \\ 0 & 1 & -1 & -5 & : & -13 \\ 0 & 1 & -3 & 0 & : & 4 \end{bmatrix} \text{ Operating } -\frac{1}{9}R_2$$

$$- \begin{bmatrix} 2 & 1 & 2 & 1 & : & 6 \\ 0 & 1 & 0 & -1 & : & -2 \\ 0 & 1 & -1 & -5 & : & -13 \\ 0 & 1 & -3 & 0 & : & 4 \end{bmatrix} \text{ Operating } R_1 - R_2, R_3 - R_2, R_4 - R_2$$

$$- \begin{bmatrix} 2 & 0 & 2 & 2 & : & 8 \\ 0 & 1 & 0 & -1 & : & -2 \\ 0 & 0 & -1 & -4 & : & -11 \\ 0 & 0 & -3 & 1 & : & 6 \end{bmatrix} \text{ Operating } R_4 - 3R_3, \frac{1}{2}R_1$$

$$- \begin{bmatrix} 1 & 0 & 1 & 1 & : & 4 \\ 0 & 1 & 0 & -1 & : & -2 \\ 0 & 0 & -1 & -4 & : & -11 \\ 0 & 0 & 0 & 13 & : & 39 \end{bmatrix} \text{ Operating } R_1 + R_3, \frac{1}{13}R_4$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & -3 & \vdots & -7 \\ 0 & 1 & 0 & -1 & \vdots & -2 \\ 0 & 0 & -1 & -4 & \vdots & -11 \\ 0 & 0 & 0 & 1 & \vdots & 3 \end{bmatrix}$$

Operating $R_1 + 3R_4, R_2 + R_4, R_3 + 4R_4$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & \vdots & 2 \\ 0 & 1 & 0 & 0 & \vdots & 1 \\ 0 & 0 & -1 & 0 & \vdots & 1 \\ 0 & 0 & 0 & 1 & \vdots & 3 \end{bmatrix}$$

Operating $(-1)R_3$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 & \vdots & 2 \\ 0 & 1 & 0 & 0 & \vdots & 1 \\ 0 & 0 & 1 & 0 & \vdots & -1 \\ 0 & 0 & 0 & 1 & \vdots & 3 \end{bmatrix}$$

Hence $x_1 = 2, x_2 = 1, x_3 = -1, x_4 = 3$.**Example 4.** Using matrix method, show that the equations :

$$3x + 3y + 2z = 1, \quad x + 2y = 4, \quad 10y + 3z = -2, \quad 2x - 3y - z = 5$$

are consistent and hence obtain the solutions for x, y and z .**Sol.** In matrix notation, the given system of equations can be written as

$$AX = B$$

$$\text{where } A = \begin{bmatrix} 3 & 3 & 2 \\ 1 & 2 & 0 \\ 0 & 10 & 3 \\ 2 & -3 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 1 \\ 4 \\ -2 \\ 5 \end{bmatrix}$$

Augmented matrix $[A : B]$

$$= \begin{bmatrix} 3 & 3 & 2 & \vdots & 1 \\ 1 & 2 & 0 & \vdots & 4 \\ 0 & 10 & 3 & \vdots & -2 \\ 2 & -3 & -1 & \vdots & 5 \end{bmatrix}$$

Operating R_{12}

$$\sim \begin{bmatrix} 1 & 2 & 0 & \vdots & 4 \\ 3 & 3 & 2 & \vdots & 1 \\ 0 & 10 & 3 & \vdots & -2 \\ 2 & -3 & -1 & \vdots & 5 \end{bmatrix}$$

Operating $R_2 - 3R_1, R_4 - 2R_1$

$$\sim \begin{bmatrix} 1 & 2 & 0 & \vdots & 4 \\ 0 & -3 & 2 & \vdots & -11 \\ 0 & 10 & 3 & \vdots & -2 \\ 0 & -7 & -1 & \vdots & -3 \end{bmatrix}$$

Operating $R_3 + 3R_2, R_4 - 2R_2$

$$\sim \begin{bmatrix} 1 & 2 & 0 & \vdots & 4 \\ 0 & -3 & 2 & \vdots & -11 \\ 0 & 1 & 9 & \vdots & -35 \\ 0 & -1 & -5 & \vdots & 19 \end{bmatrix}$$

Operating $R_1 - 2R_3, R_2 + 3R_3, R_4 + R_3$

$$\sim \begin{bmatrix} 1 & 0 & -18 & \vdots & 74 \\ 0 & 0 & 29 & \vdots & -116 \\ 0 & 1 & 9 & \vdots & -35 \\ 0 & 0 & 4 & \vdots & -16 \end{bmatrix}$$

Operating $R_{23}, \frac{1}{4}R_4$

$$\sim \begin{bmatrix} 1 & 0 & -18 & : & 74 \\ 0 & 1 & 9 & : & -35 \\ 0 & 0 & 29 & : & -116 \\ 0 & 0 & 1 & : & -4 \end{bmatrix}$$

Operating $R_1 + 18R_4$, $R_2 - 9R_4$, $R_3 - 29R_4$

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & : & 1 \\ 0 & 0 & 0 & : & 0 \\ 0 & 0 & 1 & : & -4 \end{bmatrix}$$

Operating R_{34}

$$\sim \begin{bmatrix} 1 & 0 & 0 & : & 2 \\ 0 & 1 & 0 & : & 1 \\ 0 & 0 & 1 & : & -4 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\rho(A) = \rho(A : B) = 3 = \text{number of unknowns}$$

$\Rightarrow$ The given system of equations is consistent and the solution is $x = 2, y = 1, z = -4$.

Example 5. For what values of λ and μ do the system of equations :

$$x + y + z = 6, \quad x + 2y + 3z = 10, \quad x + 2y + \lambda z = \mu$$

have (i) no solution (ii) unique solution (iii) more than one solutions ?

Sol. In matrix notation, the given system of equations can be written as

$$AX = B$$

where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$

Augmented matrix $[A : B]$

$$= \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix}$$

Operating $R_2 - R_1, R_3 - R_1$

$$\sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 1 & \lambda - 1 & : & \mu - 6 \end{bmatrix}$$

Operating $R_1 - R_2, R_3 - R_2$

$$\sim \begin{bmatrix} 1 & 0 & -1 & : & 2 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & \lambda - 3 & : & \mu - 10 \end{bmatrix}$$

Case I. If $\lambda = 3, \mu \neq 10$

$$\rho(A) = 2, \rho(A : B) = 3$$

$$\rho(A) \neq \rho(A : B)$$

$\therefore$ The system has no solution.

Case II. If $\lambda \neq 3, \mu$ may have any value

$$\rho(A) = \rho(A : B) = 3 = \text{number of unknowns}$$

$\therefore$ The system has unique solution.

Case III. If $\lambda = 3, \mu = 10$

$$\rho(A) = \rho(A : B) = 2 < \text{number of unknowns}$$

$\therefore$ The system has an infinite number of solutions.

Example 6. Solve the equations :

$$x_1 + 3x_2 + 2x_3 = 0, 2x_1 - x_2 + 3x_3 = 0, 3x_1 - 5x_2 + 4x_3 = 0, x_1 + 17x_2 + 4x_3 = 0.$$

Sol. In matrix notation, the given system of equations can be written as

$$AX = 0$$

where $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & -1 & 3 \\ 3 & -5 & 4 \\ 1 & 17 & 4 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Operating $R_2 - 2R_1, R_3 - 3R_1, R_4 - R_1$

$$A \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & -7 & -1 \\ 0 & -14 & -2 \\ 0 & 14 & 2 \end{bmatrix}$$

Operating $R_3 - 2R_2, R_4 + 2R_2$

$$\sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & -7 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Operating $R_1 + 2R_2$

$$\sim \begin{bmatrix} 1 & -11 & 0 \\ 0 & -7 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = 2 < \text{number of unknowns}$$

$\Rightarrow$ The system has an infinite number of non-trivial solutions given by

$$x_1 - 11x_2 = 0, \quad -7x_2 - x_3 = 0$$

i.e.,

$$x_1 = 11k,$$

$$x_2 = k,$$

$$x_3 = -7k, \text{ where } k \text{ is any number.}$$

Different values of k give different solutions.

TEST YOUR KNOWLEDGE

- Write the following equations in matrix form $AX = B$ and solve for X by finding A^{-1} :
 - $2x - 2y + z = 1$
 $x + 2y + 2z = 2$
 $2x + y - 2z = 7$
 - $2x_1 - x_2 + x_3 = 4$
 $x_1 + x_2 + x_3 = 1$
 $x_1 - 3x_2 - 2x_3 = 2$
- Using the loop current method on a circuit, the following equations were obtained :
 $7i_1 - 4i_2 = 12, -4i_1 + 12i_2 - 6i_3 = 0, -6i_2 + 14i_3 = 0$.
 By matrix method, solve for i_1, i_2 and i_3 .
- Solve the following systems of equations by matrix method :
 - $x + y + z = 8, \quad x - y + 2z = 6, \quad 3x + 5y - 7z = 14$
 - $x + y + z = 6, \quad x - y + 2z = 5, \quad 3x + y + z = 8$
 - $x + 2y + 3z = 1, \quad 2x + 3y + 2z = 2, \quad 3x + 3y + 4z = 1$